

# W. PABODA SATHSARANI FERNANDO

Data Science • AI/ML • Full-Stack Intern Candidate

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## PROFESSIONAL SUMMARY

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Third-year BSc (Hons) Information Technology undergraduate at SLIIT specializing in Data Science, with practical experience across full-stack web development, mobile applications, and machine learning projects. Skilled in developing secure REST APIs, data-driven systems, interactive dashboards, and AI-supported applications using Java, Python, JavaScript, React, Node.js, Spring Boot, MongoDB, MySQL, and machine learning technologies. Currently seeking a Data Science, AI/ML or Software Engineering internship to apply technical knowledge to real-world solutions and contribute effectively within a

## PROJECT EXPERIENCE

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### Fitness Tracking Mobile Application

April 2026 - May 2026

React Native (Expo) | Node.js | Express.js | MongoDB | REST APIs | JWT

- Developed a full-stack mobile fitness tracking application that helps users manage workouts, exercises, fitness goals, meal plans, weekly challenges, and progress records.
- Implemented secure authentication and protected API access using JWT, bcrypt password hashing, and role-based backend middleware for reliable user session management.
- Built and integrated RESTful APIs with MongoDB/Mongoose for core modules including User Profiles, Workouts, Exercises, Goals, Meal Plans, Challenges, and Progress Analytics.
- Designed mobile screens with React Native and Expo, including dashboard views, progress tracking, workout management, and analytics charts for a smooth user experience.

### AI-Driven Loan Risk Assessment & Recommendation System

January 2026 - May 2026

React | FastAPI | MongoDB | Python | XGBoost | SHAP

- Developed a secure AI-driven microfinance loan platform to predict applicant default risk and support data-driven loan approval decisions
- Built and integrated an XGBoost-based risk prediction model using feature engineering, SMOTE, and SHAP explanations for transparent risk assessment
- Implemented role-based access, loan application workflows, recommendation logic, repayment tracking, and analytics dashboards for staff users

### DengueRisk - AI-Based Dengue Risk Prediction System

August 2025 - October 2025

Python | Pandas | NumPy | Scikit-learn | Matplotlib | Data Visualization

- Trained a Scikit-learn classification model to predict dengue risk levels across 25 Sri Lankan districts using weekly case counts and weather data.
- Performed data preprocessing, missing-value handling, and feature engineering on multi-source datasets to improve model accuracy.
- Created visualizations of dengue trends, seasonal patterns, and risk heatmaps to support early outbreak detection and public health analysis.

### Secure Web-Based E-Voting System for Campus Awards

July 2025 - October 2025

Java | React | Vite | H2 Database | MS SQL Server

- Built a secure web-based e-voting platform enforcing one-vote-per-user-per-category for campus award nominations, with full input validation and session management.
- Developed admin-controlled event management and nomination management features with real-time result dashboards.

- Created transparent result-viewing interfaces enabling administrators and voters to track outcomes during and after voting periods.

### Real Estate Property Listings Portal

March 2025 - May 2025

Java | JavaScript | HTML | CSS | Smart Tomcat

- Developed a web-based application for managing and displaying property listings.
- Applied object-oriented programming concepts to structure system design.
- Implemented property management features with a user-friendly interface.

## TECHNICAL SKILLS

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**Programming Languages:** Java, Python, JavaScript

**Frontend Development:** React, React Native, Expo, Vite, HTML, CSS

**Backend & Frameworks:** Node.js, Express.js, REST APIs, JWT Authentication

**Databases & Data Management:** MongoDB, MySQL, MS SQL Server, H2

**Machine Learning & Data Science:** XGBoost, Scikit-learn, Pandas, NumPy, Data Visualization, SMOTE

**Core Concepts:** Object-Oriented Programming, RESTful Architecture, Agile Methodology, Database Design, Authentication & Authorization

**Tools & Technologies:** Git, GitHub, Postman, IntelliJ IDEA, VS Code, Mongoose

## SOFT SKILLS

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• Problem Solving • Analytical Thinking • Teamwork & Collaboration • Communication (Verbal & Written) • Time Management • Adaptability • Continuous Learning

## EDUCATION

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- Undergraduate – BSc (Hons) Information Technology (Data Science)  
Sri Lanka Institute of Information Technology (SLIIT)  
**Expected Graduation: 2028**
- G.C.E Advanced Level (A/L) – Physical Science Stream  
Panadura Balika Maha Vidyalaya  
**2020 – 2023**
- G.C.E Ordinary Level (O/L)  
Panadura Balika Maha Vidyalaya  
**2018 – 2019**

## EXTRA CURRICULAR ACTIVITIES

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- A Student Member of the Institute of Electrical and Electronics Engineers (IEEE) (2026 - Present)

## REFERENCES

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References available upon request.